1. What is sorting?

Answer:
Sorting is the process of arranging data in a specific order, such as ascending or descending.

2. What is Selection Sort?

Pass 1: scan the unsorted suffix

Comparison 1 of 7·0 sorted

Minimum 6Comparing 3

6

3

8

2

7

1

5

4

1

2

3

4

5

6

7

8

Compare 3 with 6; the bars stay fixed while this pass scans

Selection step

Selection step

Answer:
Selection Sort is a comparison-based sorting algorithm that repeatedly finds the

smallest element from the unsorted portion of the array and places it at its correct position.

3. What is Bubble Sort?

Compare 5 and 2 → swap

Pass 1·Comparison 1 of 6·1 swap

2

5

7

1

6

3

4

1

2

3

4

5

6

7

Result: 2, 5; 5 moves one place right

Sorting step

Sorting step

Answer:
Bubble Sort repeatedly compares adjacent elements and swaps them if they are in the wrong order. After each pass, the largest element moves to its correct position.

4. Why are floating-point numbers used for salaries?

Answer:
Salaries may contain decimal values (e.g., 45678.75), so floating-point numbers are used.

5. What is the time complexity of Selection Sort?

Answer:

- Best Case: **O(n²)**
- Average Case: **O(n²)**
- Worst Case: **O(n²)**

6. What is the time complexity of Bubble Sort?

Answer:
For the given implementation:

- Best Case: **O(n²)**
- Average Case: **O(n²)**
- Worst Case: **O(n²)**

*(With an optimized Bubble Sort using a swap flag, the best case becomes **O(n)**.)*

7. What is the space complexity of both algorithms?

Answer:
Both Selection Sort and Bubble Sort require **O(1)** extra space because they sort the array in place.

8. Which algorithm performs fewer swaps?

Answer:
Selection Sort performs fewer swaps than Bubble Sort because it swaps only once per pass.

9. Which algorithm is stable?

Answer:

- Bubble Sort is **stable**.
- Selection Sort is **not stable** in its standard implementation.

10. What is meant by an in-place sorting algorithm?

Answer:
An in-place sorting algorithm sorts the array without using significant extra memory.

11. How many passes are required for an array of size n?

Answer:
Both algorithms require **n − 1** passes.

12. Why is Bubble Sort called "Bubble" Sort?

Answer:
Because the largest element "bubbles up" to the end of the array after every pass.

13. Why is Selection Sort called "Selection" Sort?

Answer:
Because in each pass it selects the minimum element from the unsorted part and places it in the correct position.

14. Why do we use nested loops in these algorithms?

Answer:
The outer loop controls the number of passes, while the inner loop performs comparisons and swaps (or finds the minimum element).

15. What is the purpose of the variable temp?

Answer:
It is used to temporarily store a value during the swapping of two elements.

16. Why do we compare adjacent elements in Bubble Sort?

Answer:
Comparing adjacent elements allows larger elements to move toward the end of the array after each pass.

17. Why is Selection Sort not suitable for large datasets?

Answer:

Because it always performs **O(n²)** comparisons, making it inefficient for large datasets.

18. How are the top five highest salaries displayed?

Answer:
After sorting the salaries in ascending order, the program prints the last five elements in reverse order (from highest to lowest).

19. Can Bubble Sort be optimized?

Answer:
Yes. By using a **swap flag**, the algorithm can stop early if no swaps occur during a pass, making the best-case time complexity **O(n)**.

20. What are the applications of sorting?

Answer:

- Employee salary management
- Student result processing
- Banking systems
- Payroll systems
- Database management
- E-commerce product sorting

21. What is the difference between Selection Sort and Bubble Sort?

Selection Sort	**Bubble Sort**
Selects the minimum element in each pass	Compares adjacent elements
Performs fewer swaps	Performs more swaps
Not stable	Stable
Time Complexity: O(n²)	Time Complexity: O(n²)
In-place algorithm	In-place algorithm

22. Why do we display the top five salaries after sorting?

Answer:
Sorting arranges salaries in ascending order, making it easy to access the five highest salaries by reading the last five elements of the sorted list.

23. What is the advantage of sorting before finding the highest salaries?

Answer:
After sorting, the highest salaries are already at the end of the array, so they can be retrieved directly without additional searching.

24. Which sorting algorithm is better among Selection Sort and Bubble Sort?

Answer:
For the same input size, **Selection Sort** generally performs fewer swaps than Bubble Sort. However, both have **$O(n^2)$** time complexity and are mainly used for educational purposes rather than large real-world datasets.

25. Which sorting algorithms are preferred for large datasets?

Answer:
For large datasets, algorithms such as **Merge Sort**, **Quick Sort**, and **Heap Sort** are preferred because they have an average time complexity of **$O(n \log n)$**.